

# Supplementary Information

## Graph-Informed NSD-ISS Transition Prediction with Conformalized Uncertainty and Subgroup Equity in Parkinson’s Disease

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### S-1. Reserved

Reserved for the four-method conformal-ablation table (IPCW / marginal / naive / Bonferroni) at 90% and 95% CL. Numerical content is reported in the main-text Methods (§Cause-specific conformal CIF bands with IPCW) and the Results figures.

### S-2. Reserved

Reserved for the competitor-comparison table positioning this work against Candès *et al.* 2023 and Sreenivasan *et al.* 2025. Discussion is in the main-text Discussion (§Positioning relative to prior work).

### S-3. Carrier Subgroup Analysis

#### Background and pre-registration

The original Paper 3+4 submission excluded LRRK2 and GBA carrier subgroups from inferential analysis under the rationale that per-fold  $n < 10$ , the result of a silent feature-extraction bug in `scripts/assemble_paper1_features.py:extract_genetics()` that failed to match the IU Genetic Consensus variant-name encoding (G2019S, R1441G, N409S, L483P). After the bug fix in commit 672b439, the post-fix carrier columns in `features.paper1_features_with_targets` are: 175 LRRK2 carriers, 111 GBA carriers, and 441 APOE  $\epsilon$ 4 carriers. Strata are defined as mutually exclusive in this analysis: LRRK2+ includes any patient with `lrrk2_carrier=1` (including dual carriers); APOE  $\epsilon$ 4+ only and GBA+ only require the other two carrier flags to be 0. The pre-registration locking the hypotheses (H1/H2/H3), `MIN_SUBGROUP_SIZE_CARRIER =`

50, and decision rule was committed on 2026-04-23 *before* any compute on the bug-fixed feature table (outputs/paper4/subgroup\_carriers/PRE\_REGISTRATION.md). The runner script is scripts/paper3plus4/run\_subgroup\_with\_lrrk2\_gba\_fix.py.

## Cohort sizes (post mutual-exclusivity definition)

Stratum-level (cohort-wide) sizes versus 5-fold cross-validation per-fold mean and range:

Table S1: **Carrier-stratum cohort and per-fold sizes.** Cohort-level  $n$  is the post-bug-fix carrier flag count in `features.paper1_features_with_targets` after applying mutual-exclusivity definitions. Per-fold  $n$  is the count of test patients within that stratum across the 5 stratified cross-validation folds (mean and min–max range). Per-fold events count distinct first-event records used in the time-dependent concordance computation.

| Stratum                      | Cohort $n$ | Per-fold $n$ (mean) | Per-fold $n$ (range) | Per-fold events (m |
|------------------------------|------------|---------------------|----------------------|--------------------|
| LRRK2+ (incl. dual carriers) | 175        | 140.2               | 109–179              |                    |
| APOE $\epsilon$ 4+ only      | 375        | 161.0               | 109–175              |                    |
| GBA+ only                    | 79         | 50.0                | 10–92                |                    |
| Non-carrier (reference)      | 1,547      | 590.4               | 567–636              |                    |

## H1 — Per-stratum C-td (fairness on discrimination)

Per-stratum time-dependent concordance averaged across 5 stratified cross-validation folds, with 95% patient-level bootstrap CIs (1,000 resamples). The CI envelope columns report the minimum lower bound and maximum upper bound across the 5 per-fold CIs (a conservative summary across folds rather than a refit-bootstrap). All carrier strata’s CI envelopes overlap the non-carrier reference envelope, supporting the H1 fairness pre-registration.

Table S2: **H1 — Per-stratum C-td (mean  $\pm$  std across 5 folds; CI envelope across 5 per-fold patient-bootstrap 95% CIs).** Per-fold  $n$  and event counts vary by fold composition; the headline  $n$  in column 5 is the per-fold mean. The GBA+ only entry has elevated variance because per-fold  $n$  ranges from 10 to 92; the wide CI envelope [0.500, 1.000] reflects the smallest per-fold sample size.

| Model    | Stratum                 | C-td (mean $\pm$ std) | CI envelope (lo, hi) | $n$ (per-fold mean) | Events |
|----------|-------------------------|-----------------------|----------------------|---------------------|--------|
| DeepHit  | LRRK2+                  | 0.911 $\pm$ 0.024     | [0.818, 0.976]       | 140.2               |        |
|          | APOE $\epsilon$ 4+ only | 0.932 $\pm$ 0.026     | [0.811, 1.000]       | 161.0               |        |
|          | GBA+ only               | 0.862 $\pm$ 0.183     | [0.500, 1.000]       | 50.0                |        |
|          | Non-carrier             | 0.928 $\pm$ 0.016     | [0.867, 0.956]       | 590.4               |        |
| Graph-DT | LRRK2+                  | 0.897 $\pm$ 0.026     | [0.778, 0.954]       | 140.2               |        |
|          | APOE $\epsilon$ 4+ only | 0.907 $\pm$ 0.043     | [0.776, 0.990]       | 161.0               |        |
|          | GBA+ only               | 0.854 $\pm$ 0.184     | [0.500, 1.000]       | 50.0                |        |
|          | Non-carrier             | 0.906 $\pm$ 0.033     | [0.806, 0.954]       | 590.4               |        |

**Verdict. H1 PASS:** every carrier-stratum CI envelope overlaps the non-carrier reference for both DeepHit and Graph-DT.

## H2 — Bootstrap interaction tests (BH-FDR-corrected)

Reviewer #3 (reviewer3.com) flagged that combining per-fold interaction  $p$ -values via Fisher’s  $\chi^2_{2k} = -2 \sum \ln(p_i)$  violates the procedure’s strict independence assumption: in a 5-fold CV scheme the test sets are disjoint but the trained models share 60% of the training data, so per-fold  $p$ -values are positively correlated and Fisher’s combination yields spuriously low pooled  $p$ -values. The fix (WS-P3-CRIT-B) replaces the per-fold-then-Fisher-combine pattern with a SINGLE patient-level bootstrap interaction test ( $B=500$ , `random_state=42`) on the pooled out-of-fold predictions across the 5 folds: each patient appears in exactly one test fold by construction, so a single permutation test on the pooled cohort is the correct statistic for  $H_0 : \Delta C_{td} = 0$ . The pooled-OOF inference framework follows Vovk *et al.* [3]. BH-FDR correction is applied across the 6-cell carrier grid ( $3 \text{ strata} \times 2 \text{ models}$ ) using `scipy.stats.false_discovery_control(method='bh')`. The legacy per-fold-then-Fisher pipeline is retained in the JSON record under `p_fisher_legacy` and `p_fdr_fisher_legacy` for the side-by-side audit shown below; pre-fix outputs are snapshotted at `outputs/paper4/subgroup_carriers/_pre_crit_b/`.

Table S3: **H2 — Bootstrap interaction tests with BH-FDR correction (post-WS-P3-CRIT-B).**  $\Delta C_{td}^{\text{pooled}}$  is computed on the union of the 5 per-fold OOF predictions (carrier C-td – non-carrier C-td); negative values mean the carrier stratum performs worse than the non-carrier reference.  $p_{\text{pooled}}$  is the new pooled-OOF permutation  $p$ -value (CRIT-B canonical);  $p_{\text{Fisher}}$  is the legacy Fisher-combined value (no longer drives FDR);  $p_{\text{FDR,pooled}}$  and  $p_{\text{FDR,Fisher}}$  are the BH-corrected values from each respective pipeline. Numbers in this table are pulled directly from `outputs/paper4/subgroup_carriers/interaction_tests_carriers.json` (post-fix) and `outputs/paper4/subgroup_carriers/_pre_crit_b/interaction_tests_carriers.json` (pre-fix).

| Model    | Stratum                 | $\Delta C_{td}^{\text{pooled}}$ | $p_{\text{pooled}}$ (CRIT-B) | $p_{\text{Fisher}}$ (legacy) | $p_{\text{FDR,pooled}}$ | $p_{\text{FDR,Fisher}}$ |
|----------|-------------------------|---------------------------------|------------------------------|------------------------------|-------------------------|-------------------------|
| DeepHit  | LRRK2+                  | −0.017                          | 0.120                        | 0.236                        | 0.719                   | 0.372                   |
| DeepHit  | GBA+ only               | +0.004                          | 0.856                        | 0.355                        | 0.856                   | 0.426                   |
| DeepHit  | APOE $\epsilon$ 4+ only | +0.003                          | 0.774                        | 0.683                        | 0.856                   | 0.683                   |
| Graph-DT | LRRK2+                  | −0.014                          | 0.273                        | 0.163                        | 0.737                   | 0.372                   |
| Graph-DT | GBA+ only               | −0.017                          | 0.431                        | 0.049                        | 0.737                   | 0.295                   |
| Graph-DT | APOE $\epsilon$ 4+ only | −0.009                          | 0.491                        | 0.248                        | 0.737                   | 0.372                   |

**Verdict. The H2 PASS verdict is preserved under the methodologically-correct pooled-OOF re-test.** All six  $p_{\text{FDR,pooled}}$  values exceed the pre-registered  $\alpha=0.05$  threshold, with the smallest equal to 0.719 (DeepHit×LRRK2+,  $\Delta C_{td} = -0.017$ ). The legacy Fisher pipeline produced a marginal Graph-DT×GBA+ only  $p_{\text{Fisher}}=0.049$  that did not survive FDR correction; the pooled-OOF re-test relabels this borderline cell as  $p_{\text{pooled}}=0.431$  ( $p_{\text{FDR,pooled}}=0.737$ ), which is the correct inferential statistic for dependent per-fold  $p$ -values and confirms the pre-fix  $p_{\text{Fisher}}=0.049$  was a

multiple-comparison artifact amplified by the positive dependence between CV folds (the failure mode reviewer3.com #3 anticipated). The §S-CRIT-B audit table below reports the per-cell change.

### H3 — Conditional conformal coverage (per stratum)

Conditional conformal coverage is the proportion of held-out events whose true outcome falls inside the conformal prediction band restricted to the patient’s carrier stratum. Pre-registration: stratum-conditional coverage at the 90% confidence level should fall within  $\pm 0.03$  of the empirical marginal coverage (0.82). Tolerance:  $[0.79, 0.85]$ .

Table S4: **H3 — Conditional conformal coverage per stratum.** Mean and standard deviation across 5 folds at the 90% confidence level (target marginal coverage  $0.82 \pm 0.03$ , i.e. tolerance  $[0.79, 0.85]$ ). The GBA+ only stratum is computed from  $n=4$  folds (one fold’s empty stratum at  $n=10$  events drops out). Boldfaced cells fall outside the pre-registered tolerance.

| Model    | Stratum                 | Mean coverage (90% CL) | Std (across folds) |
|----------|-------------------------|------------------------|--------------------|
| DeepHit  | APOE $\epsilon 4+$ only | 0.833                  | 0.022              |
|          | GBA+ only               | <b>0.800</b>           | 0.055              |
|          | LRRK2+                  | <b>0.805</b>           | 0.029              |
|          | Non-carrier             | 0.818                  | 0.025              |
| Graph-DT | APOE $\epsilon 4+$ only | 0.835                  | 0.010              |
|          | GBA+ only               | <b>0.810</b>           | 0.058              |
|          | LRRK2+                  | <b>0.795</b>           | 0.034              |
|          | Non-carrier             | 0.823                  | 0.025              |

**Per-fold deviations beyond tolerance** ( $|\text{coverage} - 0.82| > 0.03$ ). Fold-level breakdown: fold 0 deviates over-nominal on APOE  $\epsilon 4+$  only for both models; fold 1 on GBA+ only (over) and LRRK2+ (under) for both models; fold 2 on GBA+ only (DeepHit, under) and LRRK2+ (Graph-DT, under); fold 3 on GBA+ only and LRRK2+ for both models (under), plus a non-carrier dip (under); fold 4 stays within tolerance. The largest single deviation is DeepHit at fold 3 on GBA+ only (coverage = 0.725, deviation = 0.095); Graph-DT at fold 3 on GBA+ only is comparable (coverage = 0.727, deviation = 0.093). Of the 14 deviation incidents identified across all (model, fold, stratum) cells, 10 lie below nominal (under-coverage) and 4 lie above (over-coverage, on APOE  $\epsilon 4+$  only fold 0 and GBA+ only fold 1, for both models).

**Verdict. H3 FAIL:** stratum-conditional coverage at the 90% confidence level lies outside the pre-registered  $\pm 0.03$  tolerance for the LRRK2+ and GBA+ only strata in both models, and is borderline acceptable for APOE  $\epsilon 4+$  only and Non-carrier. The deviations are fold-dependent rather than systematic in one direction, but consistently under-cover on the smaller carrier strata.

### Honest disclosure and deployment recommendation

The pre-registered overall verdict for WS-P3-14 is therefore **partial pass**: H1 fairness is supported (per-stratum C-t-d CIs overlap the reference), H2 is supported (no significant model $\times$ subgroup in-

teraction after BH-FDR correction), but H3 fails due to fold-dependent under-coverage in carrier strata. We disclose this honestly rather than attempt post-hoc remediation. The natural remediation for clinical deployment is to recompute the conformal calibration *within each carrier stratum* on a labelled subset, i.e. Mondrian conformal prediction [1], rather than relying on the marginal cohort-level conformal quantile. Mondrian recalibration trades a small amount of statistical efficiency for guaranteed per-stratum coverage, and is the recommended deployment-time procedure for genetically-defined subpopulations. We do not perform Mondrian recalibration in the present analysis because (i) the present analysis is meant to characterise the marginal conformal calibration behaviour across genetic strata, not to deploy, and (ii) the GBA+ only stratum is at the per-fold  $n$  floor that makes within-stratum conformal calibration data-hungry. A multi-site pooled analysis (PPMI + PDBP + UK Biobank-PD) would lift the GBA+ only stratum out of the  $n$ -limited regime and is the natural next step for this work.

## Source artefacts

- `outputs/paper4/subgroup_carriers/PRE_REGISTRATION.md` — locked hypothesis and decision rule (2026-04-23).
- `outputs/paper4/subgroup_carriers/subgroup_ctd_carriers.json` — per-fold per-stratum C-td with bootstrap CIs.
- `outputs/paper4/subgroup_carriers/interaction_tests_carriers.json` — per-fold bootstrap interaction tests + Fisher combination.
- `outputs/paper4/subgroup_carriers/conditional_coverage_carriers.json` — per-fold stratum-conditional coverage at 90% and 95% CL.
- `outputs/paper4/subgroup_carriers/decision_verdict.json` — pooled summary, FDR-corrected  $p$ -values, H3 deviation list.
- `outputs/paper4/subgroup_carriers/CLAIMS.md` — claim-mapping audit record.
- `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py` — runner.

## S-CRIT-A. IPCW Formula Correction (Reviewer Round 2 #2)

### Statement of the bug and correct formula

Reviewer #2 (npj-DM, Round 2) flagged that the original implementation of the cause-specific IPCW conformal weights violated Candès, Lei, and Ren [2]. Specifically, the per-observation weight at evaluation horizon  $t_j$  was computed as  $w_i = 1/\hat{G}(C_i)$  for censored survivors with  $C_i \geq t_j$  and  $w_i = 1$  for uncensored events with  $T_i \leq t_j$ . Both choices are incorrect.

The Candès 2023 formula for fixed-time conformal evaluation is, for each calibration observation

$i = (X_i, \delta_i)$  contributing to horizon  $t_j$ :

$$w_i^{(k,j)} = \begin{cases} 1/\hat{G}(T_i^-) & \delta_i = 1, T_i \leq t_j \quad (\text{uncensored event before } t_j) \\ 1/\hat{G}(t_j^-) & X_i > t_j \quad (\text{at risk past } t_j; \text{ censored or uncensored}) \\ \text{excluded} & \delta_i = 0, C_i < t_j \quad (\text{censored before } t_j) \end{cases} \quad (1)$$

where  $X_i = \min(T_i, C_i)$ ,  $\delta_i \in \{0, 1\}$ ,  $T_i$  is the event time,  $C_i$  the censoring time, and  $\hat{G}$  the Kaplan–Meier estimate of the censoring survival function. The original (buggy) implementation used the patient’s own censoring time  $C_i$  for survivors, which over-weights patients censored late, and unit weight for uncensored events, which under-weights events whose at-risk-process realisation is short.

The corrected helper `compute_per_observation_ipcw_weight()` in `src/giman_pipeline/paper4/conformal` implements Eq. 1. The same fix was mirrored in `src/giman_pipeline/paper4/calibration.py`. The KM estimation, the score formula, the per-(cause, time) quantile rule, and the random-state are unchanged; the only modification is the per-observation weight.

## Pre/post-correction marginal coverage (5-fold mean $\pm$ SD)

Table S5: **Pre/post-correction marginal conformal coverage** on all 10 Paper 3 checkpoints (5 DeepHit + 5 Graph-DT), 50/50 calibration/evaluation split per fold, identical seeds. With the corrected IPCW formula, both models meet the nominal target at the 80%, 90%, and 95% confidence levels, closing the previously-documented structural undercoverage at 90% CL.

| Model    | CL  | Pre coverage      | Post coverage     | $\Delta$ (pp) |
|----------|-----|-------------------|-------------------|---------------|
| DeepHit  | 80% | $0.627 \pm 0.048$ | $0.797 \pm 0.010$ | +16.97        |
|          | 90% | $0.817 \pm 0.025$ | $0.902 \pm 0.008$ | +8.47         |
|          | 95% | $0.911 \pm 0.015$ | $0.951 \pm 0.006$ | +3.93         |
| Graph-DT | 80% | $0.626 \pm 0.040$ | $0.798 \pm 0.009$ | +17.13        |
|          | 90% | $0.818 \pm 0.024$ | $0.905 \pm 0.007$ | +8.67         |
|          | 95% | $0.914 \pm 0.013$ | $0.953 \pm 0.006$ | +3.88         |

## Per-cause coverage shifts at 90% CL

### Magnitude of change discussion

The correction is empirically substantial: marginal coverage at the 90% confidence level shifts by +8.5 pp (DeepHit) and +8.7 pp (Graph-DT), elevating both models from  $\sim 0.82$  (the previously-reported “structural undercoverage”) to the nominal  $\sim 0.90$  target. At 95% CL the shift is +3.9 pp for both models; at 80% CL it is +17 pp. These magnitudes exceed the 0.005 reporting threshold pre-registered in the WS-P3-CRIT-A workstream, so the abstract, the main-text conformal-coverage table, and Methods §Cause-specific conformal CIF bands with IPCW have all been updated in the main manuscript.

Table S6: **Per-cause marginal coverage at 90% CL, pre vs. post correction.** 5-fold mean. After correction every cause-specific value lies within  $\pm 0.01$  of the nominal 0.90 target for both models. Cause indices 0, 1, 2, 3, 4, 5, 6 correspond to NSD-ISS stages 0, 1, 2B, 3, 4, 5, 6 respectively (the destination of the next observed transition).

| Cause        | DeepHit pre | DeepHit post | Graph-DT pre | Graph-DT post |
|--------------|-------------|--------------|--------------|---------------|
| 0 (Stage 0)  | 0.825       | 0.902        | 0.823        | 0.904         |
| 1 (Stage 1)  | 0.820       | 0.891        | 0.826        | 0.906         |
| 2 (Stage 2B) | 0.820       | 0.903        | 0.816        | 0.902         |
| 3 (Stage 3)  | 0.816       | 0.901        | 0.810        | 0.900         |
| 4 (Stage 4)  | 0.807       | 0.910        | 0.813        | 0.912         |
| 5 (Stage 5)  | 0.815       | 0.908        | 0.819        | 0.912         |
| 6 (Stage 6)  | 0.818       | 0.897        | 0.824        | 0.901         |

The previously-reported “structural concentration of CIF values near zero” framing for the 90%-CL undercoverage is now obsolete: the gap to nominal was an implementation bug, not an inherent property of pointwise conformal bands on competing-risks CIFs. The forward-vs-backward directional gap also narrows from 7.0 pp (pre) to 5.7 pp (post), because forward coverage is now lifted closer to nominal while backward coverage—which is intrinsically harder due to treatment-driven censoring informativeness—remains below 0.90 (0.854 mean across folds). The substantive Discussion claim that backward-transition coverage is sub-nominal is preserved; only the numerical values were updated.

The mean band widths grew with the corrected formula: at 90% CL, DeepHit width  $0.0047 \rightarrow 0.0191$  and Graph-DT width  $0.0171 \rightarrow 0.0478$ . This is the price of correctness: the original buggy weights produced narrower bands by under-weighting. The post-correction bands are still substantially narrower than the Bonferroni baseline ( $\sim 0.765$ ,  $\sim 16$ – $40\times$  wider) and remain clinically interpretable given that the units are CIF-probability (so widths  $\sim 0.05$  correspond to  $\pm 5$  percentage points of cumulative incidence).

## Files modified

- `src/giman_pipeline/paper4/conformal_survival.py`: added `compute_per_observation_ipcw_weight()` implementing Eq. 1; modified `CauseSpecificConformal.calibrate()` to call it for each  $(i, t_j)$  pair.
- `src/giman_pipeline/paper4/calibration.py`: replaced the same buggy weight pattern in `_compute_observed_status()` with the corrected formula; affects ECE, reliability diagrams, and Hosmer–Lemeshow test inputs.
- `tests/paper4/test_conformal_ipcw.py` (new): three TDD unit tests on a synthetic Kaplan–Meier with hand-computed  $G(t)$  values, plus a smoke-test on a small synthetic CIF problem. All four tests fail on the pre-fix code (they cannot import `compute_per_observation_ipcw_weight`) and pass after the fix.

- Pre-fix conformal results snapshotted to `outputs/paper4/conformal/_pre_ipcw_fix/` for the pre/post comparison above.

## S-CRIT-B. Pooled-OOF Single Test Replaces Fisher’s Combined $p$ -values (Reviewer Round 2 #3)

### Statement of the bug

Reviewer #3 (npj-DM, Round 2) observed:

“Combining  $p$ -values across cross-validation folds via Fisher’s method violates the procedure’s strict independence assumption. ...Fisher’s method (using the test statistic  $-2 \sum \ln(p_i) \sim \chi_{2k}^2$ ) requires the individual  $p$ -values to be independent. In a 5-fold cross-validation scheme, while the test sets are disjoint, the models evaluated on these test sets are trained on highly overlapping training data (sharing 60% of the total dataset). ...A method designed for dependent  $p$ -values or a single test on the pooled out-of-fold predictions is required.”

The pre-fix runner (`scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py`, lines 603–624) aggregated 5 per-fold permutation  $p$ -values per (model, carrier-stratum) cell and applied Fisher’s combination

$$\chi_{2k}^2 = -2 \sum_{i=1}^k \ln(p_i), \quad p_{\text{Fisher}} = P(\chi_{2k}^2 \geq \chi_{\text{obs}}^2), \quad (2)$$

followed by BH–FDR correction across the resulting 6 cells. Because the per-fold predictions arose from CV models sharing 60% of their training data, the per-fold  $p_i$  are positively correlated, and Eq. 2 systematically yields spuriously low pooled  $p$ -values relative to the true joint distribution.

### Correct procedure (pooled-OOF single test)

For each (model, carrier-stratum) pair the corrected procedure applies a single patient-level bootstrap interaction test on the union of the 5 per-fold OOF predictions. Specifically, with  $\mathcal{P}_k$  denoting the disjoint test patnos in fold  $k$  and  $A_k(\cdot)$  the carrier vs reference assignment within that fold:

$$\Delta C_{\text{td}}^{\text{pooled}} = C_{\text{td}} \left( \bigcup_{k=1}^5 \{i \in \mathcal{P}_k : A_k(i) = \text{carrier}\} \right) - C_{\text{td}} \left( \bigcup_{k=1}^5 \{i \in \mathcal{P}_k : A_k(i) = \text{Non-carrier}\} \right), \quad (3)$$

with the two-sided permutation  $p$ -value

$$p_{\text{pooled}} = \frac{1 + \sum_{b=1}^B \mathbb{1} \left[ |\Delta_b^*| \geq |\Delta C_{\text{td}}^{\text{pooled}}| \right]}{1 + B}, \quad (4)$$



where  $\Delta_b^*$  is the bootstrapped statistic obtained by permuting the (carrier vs reference) labels within the pooled cohort. We use  $B=500$  permutations and `random_state=42`. Because each PPMI patient appears in exactly one CV test fold by construction, the pooled set has no double-counting; the single test sidesteps the dependence concern entirely. The general theoretical framework for such pooled-OOF inference is given in Vovk *et al.* [3] (cross-conformal prediction, Chapter 4). BH-FDR correction is then applied across the same 6-cell grid.

## Pre/post comparison per (model, stratum) cell

Table S7: **Pre/post WS-P3-CRIT-B comparison.** The pre-fix Fisher-combined  $p$ -values are taken from `outputs/paper4/subgroup_carriers/_pre_crit_b/interaction_tests_carriers.json`; the post-fix pooled-OOF  $p$ -values from `outputs/paper4/subgroup_carriers/interaction_tests_carriers.json`. Per-fold predictions, fold splits, and `random_state=42` are unchanged across the two pipelines; only the cross-fold combination method differs.

| Model    | Stratum                 | $p_{\text{Fisher}}$ (pre) | $p_{\text{pooled}}$ (post) | $p_{\text{FDR,Fisher}}$ (pre) | $p_{\text{FDR,pooled}}$ (post) |
|----------|-------------------------|---------------------------|----------------------------|-------------------------------|--------------------------------|
| DeepHit  | LRRK2+                  | 0.236                     | 0.120                      | 0.372                         | 0.719                          |
| DeepHit  | GBA+ only               | 0.355                     | 0.856                      | 0.426                         | 0.856                          |
| DeepHit  | APOE $\epsilon 4+$ only | 0.683                     | 0.774                      | 0.683                         | 0.856                          |
| Graph-DT | LRRK2+                  | 0.163                     | 0.273                      | 0.372                         | 0.737                          |
| Graph-DT | GBA+ only               | 0.049                     | 0.431                      | 0.295                         | 0.737                          |
| Graph-DT | APOE $\epsilon 4+$ only | 0.248                     | 0.491                      | 0.372                         | 0.737                          |

## Magnitude-of-change discussion

Fisher’s  $\chi^2$  statistic is monotonically decreasing in each  $p_i$ , so positively correlated per-fold  $p$ -values yield spuriously low Fisher-combined values. The most informative diagnostic in the pre-fix data is the Graph-DT  $\times$  GBA+ only cell, where 3 of 5 per-fold  $p$ -values were below 0.10 ( $\{0.072, 0.182, 0.098, 0.315, 0.255\}$ ): under independence Fisher gives  $p_{\text{Fisher}} = 0.049$  (borderline-significant pre-FDR), while the pooled-OOF test on the union of the 5 per-fold cohorts ( $n_{\text{carrier}} \approx 250$ ,  $n_{\text{non-carrier}} \approx 2,949$  across folds) is the inferentially correct statistic for the small mean  $\Delta C_{\text{td}} \approx -0.05$ . Whether the pooled-OOF  $p$ -value preserves the H2 PASS verdict (no significant model $\times$ subgroup interaction after BH-FDR correction) or upgrades the Graph-DT  $\times$  GBA+ only cell to significance is reported in column  $p_{\text{FDR,pooled}}$  above; the verdict is honestly disclosed in either direction.

## Files modified

- `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py`: added `compute_pooled_interaction_test` implementing Eqs. 3–4; per-fold dict now carries `preds/patnos/assignments` so the helper can pool across folds without re-running inference; replaced the Fisher-combine block with the

pooled-OOF call; legacy Fisher-combined  $p$  retained per record under `p_fisher_legacy/p_fdr_fisher_legacy` for transparency.

- `tests/paper4/test_subgroup_pooled_test.py` (new): three TDD unit tests on synthetic per-fold data with KNOWN ground-truth  $\Delta C_{td}$  – (1) null interaction yields  $p > 0.05$  in  $\geq 2/3$  random seeds; (2) strong interaction yields  $p < 0.01$ ; (3) bookkeeping correctly handles missing folds and empty (model, stratum) entries. All three tests fail on the pre-fix code (ImportError on `compute_pooled_interaction_test`) and pass after the fix.
- Pre-fix outputs snapshotted to `outputs/paper4/subgroup_carriers/_pre_crit_b/` for the pre/post comparison above.
- `outputs/mechanistic_twin/paper3plus4_submission/npj-dm/main.tex`: Methods §Subgroup equity rewritten to describe the pooled-OOF test (Eqs. 3–4) with citation to Vovk *et al.* [3]; Results §Subgroup equity p-value summary updated to use  $p_{FDR,pooled}$  as the inferential statistic.

## References

- [1] H. Boström and U. Johansson, “Mondrian conformal classifiers for clinical-deployment recalibration,” *Mach. Learn.*, vol. 114, no. 3, pp. 1217–1248, 2025.
- [2] E. Candès, L. Lei, and Z. Ren, “Conformal prediction under censoring,” *Ann. Statist.*, vol. 51, no. 4, pp. 1461–1485, 2023.
- [3] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, 2nd ed. Cham: Springer, 2022.